Leo Yao

Palo Alto, CA   leoyao@andrew.cmu.edu   +1 (650) 285-0101   leowo   LeoY20

Education

Carnegie Mellon University August 2024 – May 2027
B.S. in Computer Science and Statistics & Machine Learning Pittsburgh, PA

- **QPA:** 3.68/4.00
- **Relevant Coursework:** Computer Systems, Computer Security, Theoretical Computer Science, Functional Programming, Machine Learning, Imperative Computation (Data Structures & Algorithms)
- **Awards:** Dean's List, High Honors (Spring 2025, Fall 2025)

Experience

Teaching Assistant, Principles of Functional Programming (15-150) January 2026 - Present
Carnegie Mellon University Pittsburgh, PA

- Graded 300+ student submissions in SML, providing technical feedback on correctness, efficiency, and style of recursive, parallel, and higher-order algorithms.
- Led weekly labs for 20+ students and collaborated with course staff on assignment design and grading.

Research Assistant, CERN CMS Experiment September 2025 - Present
Carnegie Mellon University Pittsburgh, PA

- Built a training and evaluation pipeline in PyTorch and PyG for quantization-aware graph neural networks, processing particle physics datasets from raw ROOT files through feature extraction and graph construction.
- Analyzing CERN HGCAL detector data to prepare adaptation of transformer-based clustering models from synthetic to real particle physics data.

Research Assistant, SPICE Lab June 2025 – August 2025
Carnegie Mellon University Pittsburgh, PA

- Engineered a data preprocessing pipeline using R Tidyverse, Pandas, NumPy, and SciKit-Learn to clean and normalize large-scale energy consumption datasets.
- Developed a neural network with PyTorch to forecast households' annual cooling energy, engineering a 75% accuracy improvement by implementing LightGBM-based feature selection and hyperparameter tuning.

Projects

PharmaSentinel | *Python, React, TypeScript, Supabase, OpenAI API* February 2026

- Orchestrated a multi-agent pipeline using Python, OpenAI API, and Brave Search MCP to monitor inventory, analyze supply chain news, and query openFDA data to detect drug shortages and identify substitutes.
- Designed PostgreSQL triggers and Supabase Realtime listeners to detect inventory changes, re-run the agent pipeline, and push shortage alerts, resolutions, and draft purchase orders to a live dashboard.

Computer Systems | *C, POSIX threads* May 2025 – August 2025

- **Dynamic Memory Allocator**: Implemented a malloc/free replacement using segregated free lists and coalescing, achieving 74.3% utilization and 15,885 kops/sec throughput (top of class).
- **Tiny Linux Shell**: Engineered a Linux shell featuring foreground/background job control, I/O redirection, and robust signal handling to manage concurrent process execution.
- **Shark File System**: Developed a thread-safe concurrent file system with standard features (e.g. file renaming, file descriptor position management), optimizing multithreaded performance and correctness with mutex synchronization.

Skills

Programming Languages: Python, C, SQL, SML, OCaml, Dafny, x86-64 Assembly, R
Frameworks/Libraries: Pandas, NumPy, scikit-learn, PyTorch, OpenAI API, Supabase, MCP
Developer Tools: Git, Linux, GDB, Valgrind